# SIAMIS ICT Weekly Lesson Plan

Teacher: Ray Joshua Balingkit | Year 10–11 | Term 2 | Week 9

Year 10 course sequence: Units 1–3. Two 45-minute lessons. Year 11 progression: Units 4–6.

## Lesson overview

This week covers risks to data: malware, accidental loss and data theft and authentication, encryption and protective software.

**Pearson Edexcel International GCSE (9-1) ICT Student Book, printed pages 96-102**

## End of week assessment

Check accurate terminology, explanations and application through the relevant book activities and questions. Record misconceptions and use the next feedback or consolidation session for corrections. Assess only content already taught.

## Lesson 1

**Risks to data: malware, accidental loss and data theft**

Date: 1 March 2027 | Duration: 45 minutes

## Learning objective

By the end of this lesson, students will be able to explain and apply risks to data: malware, accidental loss and data theft, using accurate textbook terminology and a justified example.

## Lesson flow

**Hook 5 minutes**

Use a short scenario, image or retrieval question to introduce risks to data: malware, accidental loss and data theft.

**Introduce 8 minutes**

Teach risks to data: malware, accidental loss and data theft from printed pages 96-102; include the relevant definitions, diagrams and examples.

**We do 10 minutes**

Work through a textbook example or comparison for risks to data: malware, accidental loss and data theft; explain each decision and check understanding.

**You do 17 minutes**

Complete the relevant textbook activities and questions for risks to data: malware, accidental loss and data theft. Use a practical demonstration or scenario where appropriate; finish remaining practice as homework.

**Check 5 minutes**

Explain one key idea about risks to data: malware, accidental loss and data theft and justify an answer; record misconceptions for the next lesson.

## Differentiation

Support: Provide labelled diagrams, vocabulary, a structured response frame and a worked example. Apply agreed access arrangements during independent assessment without supplying answers.

Extension: Justify an alternative solution, explain a limitation or apply the same concept to a new scenario using evidence.

## Lesson 2

**Authentication, encryption and protective software**

Date: 3 March 2027 | Duration: 45 minutes

## Learning objective

By the end of this lesson, students will be able to explain and apply authentication, encryption and protective software, using accurate textbook terminology and a justified example.

## Lesson flow

**Hook 5 minutes**

Use a short scenario, image or retrieval question to introduce authentication, encryption and protective software.

**Introduce 8 minutes**

Teach authentication, encryption and protective software from printed pages 96-102; include the relevant definitions, diagrams and examples.

**We do 10 minutes**

Work through a textbook example or comparison for authentication, encryption and protective software; explain each decision and check understanding.

**You do 17 minutes**

Complete the relevant textbook activities and questions for authentication, encryption and protective software. Use a practical demonstration or scenario where appropriate; finish remaining practice as homework.

**Check 5 minutes**

Explain one key idea about authentication, encryption and protective software and justify an answer; record misconceptions for the next lesson.

## Differentiation

Support: Provide labelled diagrams, vocabulary, a structured response frame and a worked example. Apply agreed access arrangements during independent assessment without supplying answers.

Extension: Justify an alternative solution, explain a limitation or apply the same concept to a new scenario using evidence.

## Materials

Teacher computer and display

School-approved applications, class save folder and fictional scenarios

Relevant textbook diagrams, activities and chapter questions

Student notebooks and a teacher assessment checklist

## Teacher reflection

What worked well? _____

What needs reteaching or adjustment? _____